

Clinical Trial Subject Recruitment Engine

Lab 1: Requirements Engineering & UML Use-Case Modelling — Consolidated Submission

Problem Statement #18 | Healthcare & Telemedicine | Actors: Trial Candidate, Clinical Researcher

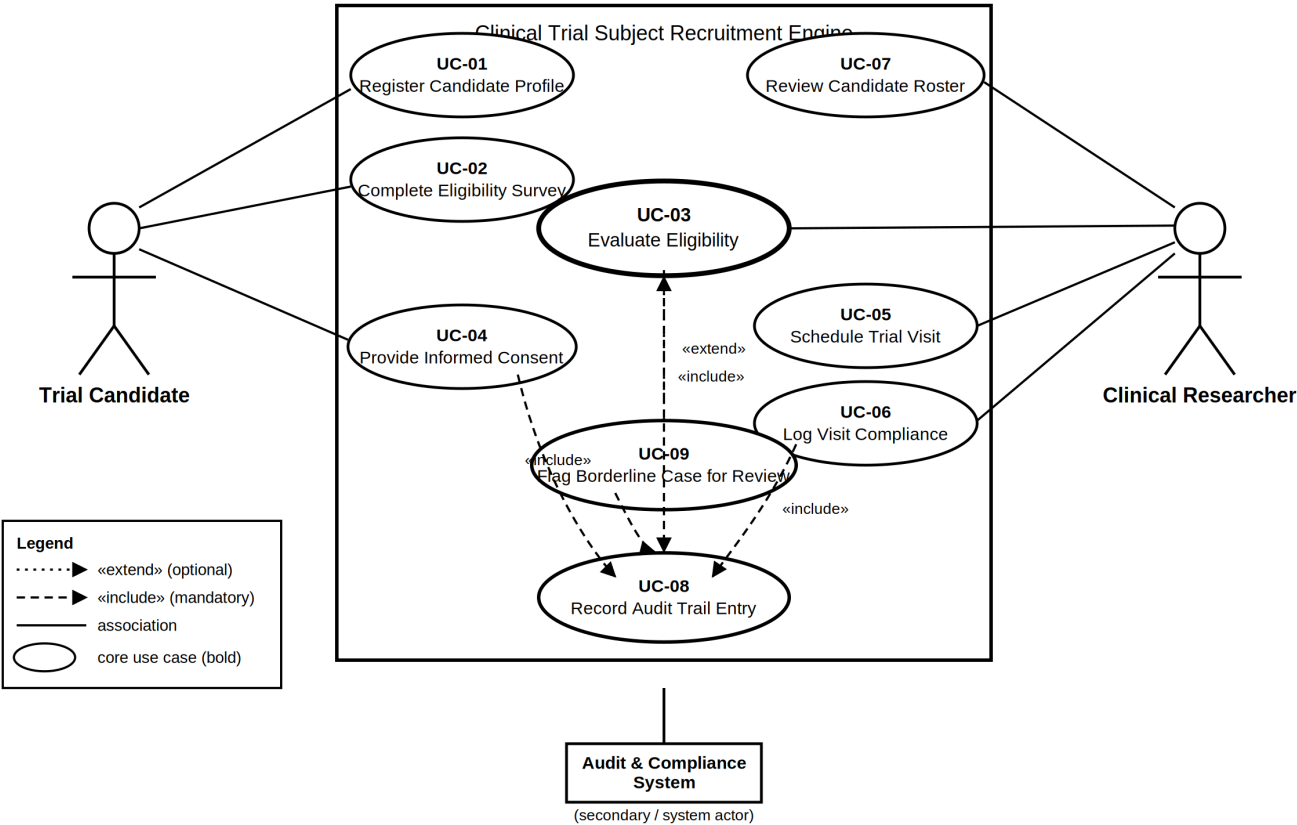
1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall evaluate a trial candidate's survey responses against the active trial protocol's inclusion/exclusion criteria and classify the candidate's eligibility status.	High	Pass: An ineligible candidate is flagged with the specific failed criteria. Fail: A candidate who meets an exclusion criterion is marked eligible.	Given in problem statement; core screening function of the platform.
FR-002	Functional	The system shall capture and timestamp each informed consent milestone (document version presented, e-signature, witness where required) for every trial candidate.	High	Pass: A consent record is created containing candidate ID, consent document version, e-signature hash, and a server timestamp within 2 seconds of signing. Fail: A consent event is stored without a version reference or timestamp.	Verifiable, auditable consent is a regulatory requirement (Good Clinical Practice / 21 CFR Part 50) before any protocol procedure begins.
FR-003	Functional	The system shall log each scheduled and completed trial visit against the protocol's visit schedule and automatically flag visits completed outside the permitted visit window.	Medium	Pass: A visit logged more than 3 days outside its protocol window is automatically tagged "Out of Window". Fail: An out-of-window visit is logged with no flag raised.	Visit-window deviations can invalidate collected trial data and must be tracked for protocol compliance reporting.
FR-004	Functional	The system shall allow a Clinical Researcher to view, filter, and search candidates by eligibility status, consent status, and visit-compliance status from a single recruitment dashboard.	Medium	Pass: Filtering by "Ineligible" returns only candidates with at least one failed inclusion/exclusion criterion. Fail: The filtered result set includes any eligible candidate.	Researchers must efficiently triage a growing recruitment pipeline without querying raw data.
FR-005	Functional	The system shall route a candidate whose survey response falls within a configurable borderline threshold of an exclusion criterion to a manual-review queue instead of automatically rejecting them.	Medium	Pass: A response scoring within 5% of an exclusion threshold is routed to the "Pending Manual Review" queue rather than auto-rejected. Fail: Such a candidate is auto-rejected without reaching the queue.	Rigid automated rejection risks false-negative exclusion of otherwise-eligible candidates near a threshold.
NFR-001	Nonfunctional (Performance & Security)	The recruitment database must comply with 21 CFR Part 11 electronic record standards, including cryptographic audit trails, for all eligibility and consent records.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load. Fail: Audit trail entries can be altered without detection, or latency exceeds target under peak load.	Given in problem statement; regulatory compliance is mandatory for any electronic clinical trial record.
NFR-002	Nonfunctional (Availability & Reliability)	The recruitment platform shall maintain at least 99.5% uptime during active enrollment periods, with automatic failover of the eligibility-evaluation service.	High	Pass: Monthly uptime reports show $\geq 99.5\%$ availability and failover completes within 30 seconds of a primary-node failure. Fail: Measured downtime exceeds 0.5% in a month, or failover exceeds 30 seconds.	Continuous access to enrollment and screening is critical to meeting trial recruitment timelines.

Total: 5 Functional Requirements (FR-001 – FR-005), 2 Nonfunctional Requirements (NFR-001 – NFR-002).

2. UML Use-Case Diagram

Clinical Trial Subject Recruitment Engine — Use-Case Diagram



3. Use-Case Flow Specification

Core Use Case: UC-03 Evaluate Eligibility

Use Case	UC-03: Evaluate Eligibility
Primary Actor	Trial Candidate (submits survey); Clinical Researcher (receives result / handles manual review)
Trigger	Trial Candidate submits a completed eligibility survey.
Preconditions	1) The Trial Candidate has a registered profile. 2) The eligibility survey has been fully completed. 3) The active trial protocol's inclusion/exclusion criteria are loaded into the system.
Postconditions	1) The candidate's eligibility status (Eligible / Ineligible / Pending Manual Review) is recorded. 2) A cryptographic audit-trail entry is created (21 CFR Part 11). 3) The Clinical Researcher's dashboard reflects the updated status.

Main Success Scenario

1. The Trial Candidate submits the completed eligibility survey through the recruitment portal.
2. The system retrieves the active trial protocol's inclusion/exclusion criteria.
3. The system evaluates each survey response against the protocol criteria in sequence («include» Record Audit Trail Entry, invoked per check performed).
4. The system computes an overall eligibility classification for the candidate.
5. The system records the classification together with the specific criteria matched or failed.
6. The system creates a cryptographic, timestamped audit-trail entry recording the protocol version, inputs evaluated, and outcome («include» Record Audit Trail Entry).
7. The system updates the Clinical Researcher's dashboard with the candidate's new eligibility status.
8. The use case ends successfully with the candidate's status recorded as “Eligible” or “Ineligible.”

Alternate Flow — 3a: Borderline Response Detected

- 3a1.** While evaluating a response, the system detects a value within a configurable borderline threshold of an exclusion criterion («extend» Flag Borderline Case for Manual Review, from UC-03).
- 3a2.** The system sets the candidate's status to “Pending Manual Review” and adds the candidate to the Clinical Researcher's review queue.
- 3a3.** The Clinical Researcher opens the candidate record and manually reviews the borderline response against protocol intent.
- 3a4.** The Clinical Researcher confirms the final eligibility decision (Eligible / Ineligible).
- 3a5.** The system records the researcher's decision as the final eligibility classification and resumes the main flow at step 5.